

**DEPARTMENT OF ECONOMICS
UNIVERSITY OF ST ANDREWS BUSINESS SCHOOL
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UNDERGRADUATE RESEARCH PROJECT IN ECONOMICS

**TITLE: Do Clubs Take Financial Risks to Secure
Promotion or Avoid Relegation? Evidence from the
EFL Championship and 2. Bundesliga**

I hereby certify that this project, which is approximately **7,700** words in length, has been composed by me, that it is the record of work carried out by me and that it has not been submitted in any previous application for a degree.

This project was conducted by me at the University of St Andrews from **January 2026** to **April 2026** towards fulfilment of the requirements of the University of St Andrews for the degree of **Economics BA (International Honours)** under the supervision of **Ian Smith**.

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Introduction

European professional football presents one of the most puzzling paradoxes in economics. Clubs routinely overspend on player wages, accumulate losses, and flirt with insolvency, yet continue to operate, attract investment, and survive. This project examines the two most commercially and competitively significant second divisions in European football, England's EFL Championship and Germany's 2. Bundesliga, to investigate whether clubs exhibit the financially irrational yet strategically rational behaviour of overspending on wages relative to revenue and profit growth. Evans, Walters, and Hamil (2022) term this practice 'financial gambling.'

Understanding the reasons for this behaviour is important as the incentive structures differ sharply between the two leagues. In England, the revenue gap between the EFL Championship and the Premier League is vast: an EFL Championship club generates roughly one-seventh of a Premier League club's revenue in the 2023/24 season (SBI Barcelona, 2026). Additionally, parachute payments distort the cutoff point between the English divisions. In Germany, the earnings gap is consequential but significantly smaller than the English gap, promotion routes are less varied, and the near-universal 50+1 ownership structure restricts the market from the types of investors willing to absorb sustained losses in pursuit of sporting success. These institutional differences make the two second divisions a compelling comparative case.

This project adopts the financial gambling classifier of Evans et al. (2022) and evaluates outcomes through a proposition-based payoff matrix, with a distinct English and German matrix due to different league and incentive structures, across the 2021/22 to 2023/24 seasons. The central hypothesis is that financial gambling is more prevalent in the EFL Championship than in the 2. Bundesliga, owing to stronger promotion incentives and

fewer ownership constraints. The findings support this, but the more significant contribution lies in the uncovering of a distributional pattern: gambling in the EFL Championship clusters among promotion-chasing clubs, while in the 2. Bundesliga it appears almost exclusively among relegation-threatened clubs.

Literature Review

Professional football has attracted much economic scholarly attention in the last 30 years, yet there is still an unresolved tension between sporting ambition and financial sustainability. This literature review synthesises key contributions, both theoretical and empirical, that frame the central question of this project: do clubs in Germany's 2. Bundesliga exhibit the same pattern of financially irrational yet strategically rational "financial gambling" behaviour Evans, Walters, and Hamil (2022) identify in the EFL Championship (hereafter Championship). The project also investigates whether the DFL's licensing and governing scheme acts as a source of "regulatory legitimacy" rather than a genuinely disruptive force against overspending, as Evans et al. (2022) also outline.

Sloane (1971) establishes the theoretical framework for understanding football club behaviour. Observing that clubs systematically overspend on players, Sloane argues that British football clubs are better understood as utility maximisers rather than profit maximisers. Unlike profit-maximising firms, clubs maximise sporting success, which generates the greatest utility for all stakeholders. This makes player overspending rational even when it produces losses. Lago, Simmons, and Szymanski (2006) extend this to the European scale, diagnosing chronic deficit financing as a systemic feature of the football industry rather than the condition of individual clubs. Counterintuitively, the authors identify rising TV revenue as a key source of instability, as clubs extrapolate from growing broadcast

income to justify wage commitments that their underlying finances cannot sustain. This dynamic has only intensified since 2006.

Storm and Nielsen (2012) formalise the theoretical mechanism linking these incentive structures to observable financial distress, applying Kornai's (1980) insight that organisations insulated from insolvency by external sponsors lose any meaningful incentive to balance their books. Examining European top divisions, the authors argue that the institutional environment virtually guarantees clubs' sustained survival regardless of financial performance. Though limited to first divisions, the framework implies that regulation alone cannot resolve chronic loss-making. If the regulatory architecture reinforces the behaviour it purports to prohibit, rules become self-defeating.

This paradox is central to Evans et al.'s (2022) study of "gambling" in the EFL Championship. The authors' definition of this gambling behaviour is that clubs intentionally overspend on wages relative to revenue and profit growth. Using data from club financial statements across 15 seasons from 2004/05 to 2018/19, their study quantifies the prevalence of this gambling behaviour and finds it is widespread and the most successful route to achieving promotion for non-relegated clubs. Additionally, the authors introduce the concept of "regulatory legitimacy," which builds on concepts already explored in other papers in this literature review. They argue that a regulatory framework itself confers legitimacy on overspending, rendering rules ineffective even when implemented. The EFL introduced the Profitability and Sustainability Rules (P&S) during Evans et al.'s study (first introduced in the 2012/13 season), and found no meaningful change in the prevalence of gambling before and after the financial sustainability rules were introduced. The EFL's P&S thus operate as a socially sanctioned permission structure rather than a financial constraint. This finding has

direct implications for any second division where a regulatory regime exists but structural promotion incentives remain overwhelming.

Wilson, Ramchandani, and Plumley (2018) demonstrate that clubs relegated from the Premier League who receive Parachute Payments have a 40% chance of immediate promotion and accrue, on average, five more points than clubs without parachute payments. The authors ultimately conclude that parachute payments have seriously eroded the competitive balance of the Championship. Parachute payments are financial distributions made by the Premier League to newly relegated clubs over a two-to-three-year period. These payments cushion the revenue shock of dropping out of the top division. Since Germany has no equivalent, this asymmetry is central to understanding the divergence in financial behaviour between the two leagues. For Championship clubs, this dissonance, both between divisions and between recipients and non-recipients of parachute payments, defines the competitive and financial environment they compete in. Plumley, Serbera, and Wilson (2021) reinforce this, finding that the majority of Championship clubs are in significant financial distress according to standard bankruptcy prediction indicators, even after the EFL introduced the sustainability rules. Together, this body of English evidence establishes the empirical standard against which this project measures the 2. Bundesliga's financial behaviour.

The 2. Bundesliga provides the comparative axis for this project. Historically, the literature on German football has projected a markedly different narrative. Frick and Prinz (2006) find that in the early 2000s, Bundesliga club revenues increase and liabilities remain low, arguing that the crisis narrative, which had been prevalent at the time, was due to the exceptional circumstances of two large clubs rather than structural forces. The paper's legacy is that it remains the reference point for the "German exceptionalism" premise, which attributes financial stability to the DFL's licensing structure and the famous 50+1

member-association ownership model. This model requires that a registered members' association retain a majority of voting rights, preventing external investors from acquiring controlling stakes. However, Szymanski and Weimar (2019) directly challenge this consensus by demonstrating that, when the entire footballing pyramid in Germany is inspected, they find that the proportion of insolvency rates in the lower German professional divisions is comparable to those in England and France. Notably, insolvency cases concentrate from the third tier and down in Germany, indicating that relegation from the 2. Bundesliga carries harsher consequences than relegation from the Bundesliga. The authors' findings contradict the notion of German exceptionalism and open the door for this project's investigation.

The distinctive ownership framework of German football introduces a significant layer of complexity to the project. Dietl and Franck (2007) argue that the lack of strict regulatory enforcement by the DFL, in combination with the member-associations system, leads to a soft budget constraint problem. This combination produces a uniquely German form of moral hazard and structural overinvestment in players. Counterintuitively, Budzinski and Kunz-Kaltenhäuser (2025) find that the 50+1 ownership structure, often heralded as a safeguard against the profligate ownership culture prevalent in the UK, has actually exacerbated financial and competitive disparities within the German football structure. Overall, the research suggests that the incentive for risk-taking behaviour is just as potent in Germany as in England, the difference lying merely in the institutional form this risk-taking adopts.

Taken together, this body of literature establishes the theoretical and empirical foundation for this project. The core concepts of utility maximisation, regulatory legitimacy, and soft budget constraints explain financial risk-taking irrespective of national setting, while Evans et al. (2022) provide the analytical blueprint this project replicates. The distinctive

German environment, specifically the 50+1 ownership model and the absence of parachute payments, distinguishes this project from a simple replication. It presents a theoretically significant test: is regulatory legitimacy confined to the English governance context, or is it a structural feature of second-tier professional football generally, wherever strong promotion and relegation incentives exist? The remainder of this project addresses this central question.

Methodology

This project employs an analytical framework adapted from Evans et al. (2022) to assess both the outcomes and prevalence of “financial gambling” among clubs in the Championship and the 2. Bundesliga. The methodology does not employ any conventional econometric approaches, but instead relies on a combination of classification rules for identifying clubs that are gambling and a payoff matrix framework for evaluating the outcomes of that behaviour. This approach is well-suited to the central question of the project, as it assesses clubs directly and more efficiently according to a club’s perceived ambition.

For this project, in both the German and the English contexts, a club is classified as gambling in a given season if, and only if, it satisfies all three of the following conditions simultaneously.

1. The club’s wage expenditure must have increased in the current season relative to the prior season.
2. The club’s wage expenditure must have increased by more than the revenue increase over the same period.
3. The club’s wage expenditure must have increased by more than the change in profit over the same period.

This three-pronged criterion ensures that the classification captures clubs that are deliberately increasing wage spending relative to their financial capacity. A club that increases wage spending in line with turnover growth would not qualify as a gambler under this definition, since such behaviour would be financially sustainable. Only when wage growth exceeds both revenue growth and profit change is a club deemed to have engaged in financial gambling..

Building on this, clubs are not assessed against a uniform benchmark but are instead assigned to propositions based on their finishing position in the prior season. The rational incentive to gamble financially will differ significantly depending on a club's ambition, proxied here by its finishing position in the prior season. Applying a uniform benchmark across all clubs would be misleading. The proposition structure groups clubs by the objectives they could reasonably pursue given their prior finish. For the Championship, four propositions are specified:

- Proposition I covers clubs finishing 3rd to 6th in the prior season, whose primary objective in the subsequent season is assumed to be securing automatic promotion, or to win promotion via a play-off position.
- Proposition II covers clubs finishing 7th to 12th, who may increase wage spending to target at least a play-off finish.
- Proposition III covers clubs finishing 13th to 18th, who may gamble either to push for a play-off place or to avoid relegation.
- Proposition IV covers clubs finishing 19th to 21st, whose dominant incentive is assumed to be avoiding relegation.

Clubs finishing 1st or 2nd and the play-off winners are excluded since they were promoted, as well as clubs finishing 22nd to 24th, as they were relegated. This project also

excludes clubs newly relegated from the Premier League or promoted from League One, since these clubs have not spent two consecutive seasons in the Championship and carry materially different financial structures.

The 2. Bundesliga has a different league structure to the Championship: the league consists of 18 clubs as opposed to the Championship's 24, and has two automatic promotion places with one promotion play-off entrant, and on the other end of the league has two automatic relegation places and one relegation play-off entrant; for the play-offs in the German context, it is a two-legged tie against the 16th-placed Bundesliga club (promotion play-offs), or against the 3rd-placed 3. Liga club (relegation play-offs) from the prior season. For the 2. Bundesliga, three propositions are specified:

- Proposition I covers clubs finishing 3rd to 6th who are targeting automatic promotion.
- Proposition II covers clubs finishing 7th to 11th who aim to challenge for promotion places.
- Proposition III covers clubs finishing 12th to 16th who are primarily focused on avoiding relegation.

For each identified gambler, the payoff matrix plots the combination of their prior-season finishing position (which ties them to a specified proposition) and their current-season outcome to one of six zones, labelled A-F. The matrices for each league are presented in Tables 1 and 2 below, along with the zone key for each division. The zone that a club finds itself in reflects the degree to which its gamble paid off, given its specific proposition.

Table 1: Payoff Matrix for Financial Gambling in the EFL Championship

Proposition	Championship position previous season	Outcome in current season						
		1st or 2nd	Play-off winner	3rd to 6th*	7th to 12th	13th to 18th	19th to 21st	22nd to 24th
I	3rd to 6th*	A	A	E	E	E	E	F
II	7th to 12th	A	A	D	E	E	E	F
III	13th to 18th	A	A	D	D	C	C	F
IV	19th to 21st	A	A	B	B	B	B	F

* Excludes the play-off winner, who was promoted. Source: Adapted from Evans et al. (2022).

Zone	Interpretation
A	Gambling clearly paid off as club promoted.
B	Gambling clearly paid off as club avoided relegation having previously finished in the 19th to 21st positions.
C	Gambling paid off for clubs previously finishing in the 13th to 18th positions if their ambition was to avoid relegation.
D	Gambling did not pay off in the current season as although the club did progress to a higher group it failed to achieve promotion.
E	Gambling did not pay off as club failed to progress to a higher group.
F	Gambling clearly did not pay off as club relegated.

Table 2: Payoff Matrix for Financial Gambling in the 2. Bundesliga

Proposition	2. Bundesliga position previous season	Outcome in current season						
		1st or 2nd	3rd	4th to 6th	7th to 11th	12th to 15th	16th	17th to 18th
I	3rd to 6th*	A	D	E	E	E	E	F
II	7th to 11th	A	D	D	E	E	E	F
III	12th to	A	B	B	C	C	E	F

	16th**							
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* Excludes the promotion play-off winner. ** Excludes the relegation play-off loser. Source: Adapted from Evans et al. (2022).

Zone	Interpretation
A	Gambling clearly paid off, as the club achieved automatic promotion or, where relevant, at least a play-off place.
B	Gambling paid off by clearly improving league position and avoiding relegation danger.
C	Gambling paid off for clubs previously finishing in the 12th to 15th positions if their ambition was to avoid relegation.
D	Gambling partially paid off, as the club improved but did not fully achieve its apparent objective.
E	Gambling did not pay off, as the club failed to achieve its apparent objective.
F	Gambling clearly did not pay off as club relegated.

Zone A represents the clearest positive payoff for both divisions: the club is automatically promoted. By contrast, zone F represents the starkest negative payoff: despite gambling, the club is automatically relegated. Zones B and C represent positive outcomes, as these are outcomes where gambling has helped a club to achieve its ambitions to varying degrees. For example, a club finishing 19th to 21st place in the Championship that avoids relegation by gambling (Zone B in the Championship) is a clear success, while a club previously in lower mid-table that stabilises its Championship position would also consider it a success (Zone C in the Championship). Zones D and E represent failure to achieve the implied objective. Zone D indicates partial progress, while Zone E represents no meaningful improvement relative to their prior position. This structure allows for a nuanced reading of results rather than a binary success-or-failure classification, and ensures that the benchmark for success is calibrated to each club's realistic ambitions rather than to a single universal standard.

Data

The empirical analysis draws on club-level financial data for the EFL Championship and 2. Bundesliga covering three seasons: 2021/22, 2022/23, and 2023/24. This project selects this period for two reasons. Firstly, it represents the financial normalisation phase after the COVID-19 shock, a phase in which matchday revenues had largely recovered, making financial statements more comparable across clubs and seasons than those from the disrupted COVID-19 period. Second, it is the most recent window for which the majority of clubs in both divisions have filed complete accounts.

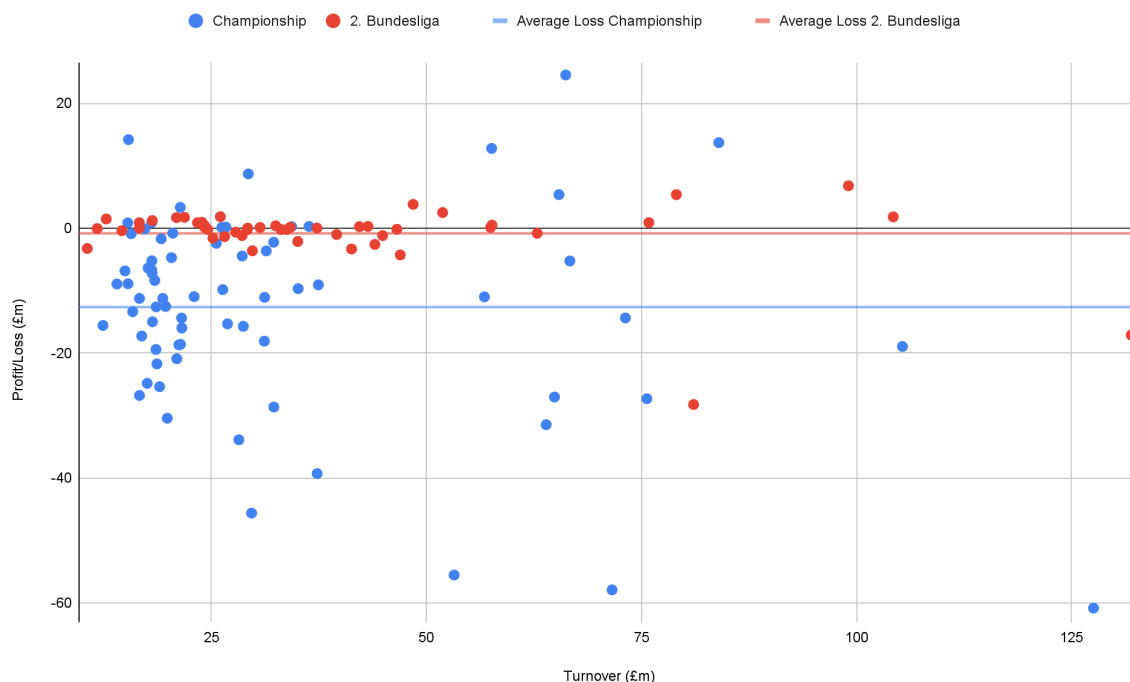
This project obtains Championship financial data from Companies House, the United Kingdom's public registry of company accounts. English clubs at this level must file annual accounts under the Companies Act; this project extracts wages, turnover, and profit/loss directly from these filings. This project sources 2. Bundesliga financial data from the Bundesanzeiger, the official federal gazette in Germany, from the annual DFL financial reports, or from club archives available on club websites. A limitation applies to German clubs structured as registered associations, "eingetragener Verein" or e.V., which are not required to publish full accounts in the Bundesanzeiger. Where this is the case, the DFL economic report's club-level figures are used as a supplementary source. Though the accounts are not exhaustive, they provide the key variables necessary for this analysis. This project takes league finishing positions from Transfermarkt (Transfermarkt, n.d.).

Total turnover, wages and salaries cost, and net profit/loss are the three financial variables used in this analysis. A fourth variable, the Wages-to-Turnover Ratio (WTR), is derived through the division of wage cost by total turnover, expressed as a percentage. UEFA (2023) commonly cites a WTR above 70% as the threshold for financial unsustainability, and this benchmark is used as a reference point throughout the analysis. All monetary values for

German clubs are converted to Great British pounds sterling using the average GBP/EUR exchange rate for the relevant season, sourced from OFX (UK), to enable cross-league comparisons. Figures are not inflation-adjusted, since the gambling classifier relies on within-club year-on-year changes rather than cross-club absolute comparisons, and the sample covered is only three years.

The financial structure of the two leagues is shown in Figure 1, which shows the profit or loss made by each club relative to their turnover per season. Championship clubs show a diverse range of turnovers, partly because recently relegated Premier League clubs receiving parachute payments sit alongside smaller clubs earning significantly less. Most Championship clubs make losses, averaging over £12 million, with some clubs recording losses exceeding £30 million. Conversely, the clubs in the 2. Bundesliga show a tight cluster around the breakeven point, regardless of each club's revenue scale.

Figure 1: Profit/Loss vs Turnover for Championship and 2. Bundesliga Clubs



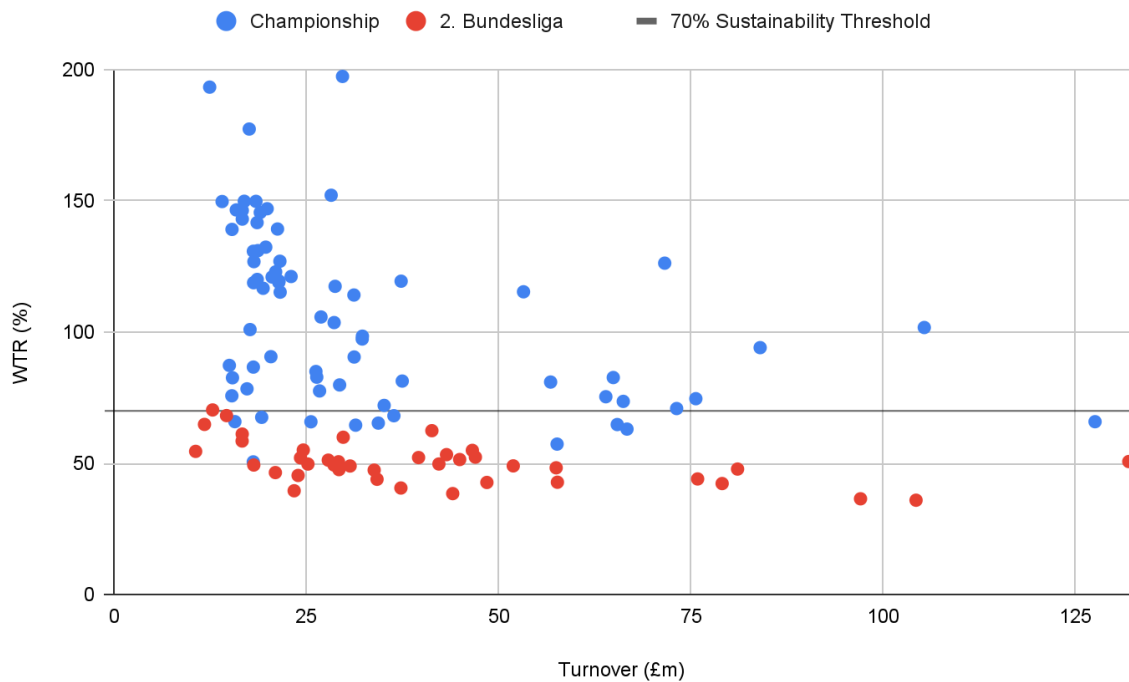
Source: Author's calculations using club financial statements.

Two data limitations apply. First, the requirement for two consecutive seasons of financial data means clubs promoted or relegated at the boundaries of the sample period are excluded. Second, financial data are unavailable for some club-seasons.

Results

Before turning to the gambling classification findings, it is worth examining the descriptive financial data in Figure 2, which clearly demonstrate the existence of a sharp distinction between the two leagues, which provides important insights into all future discussions. Figure 2 shows WTR versus turnover for each club-season in the sample. The 70% sustainability benchmark, above which wage spending is normally deemed financially unsustainable, is indicated by the black dashed horizontal line. Championship clubs cluster heavily above this benchmark, with many clubs showing WTRs higher than 100%, while several exceed 150%. These extreme values are concentrated among low-revenue clubs, i.e., those earning less than £25 million per year, implying that smaller Championship clubs are most susceptible to wage overspending relative to their revenue. In the case of the 2. Bundesliga clubs, however, the reverse is true. All but one 2. Bundesliga club fall below the 70% sustainability threshold throughout the entire revenue range, with most lying between 40% and 65%. This is consistent with the hypothesis that German clubs, operating under stricter DFL financial regulations and a different ownership culture, maintain greater wage discipline regardless of their competitive position.

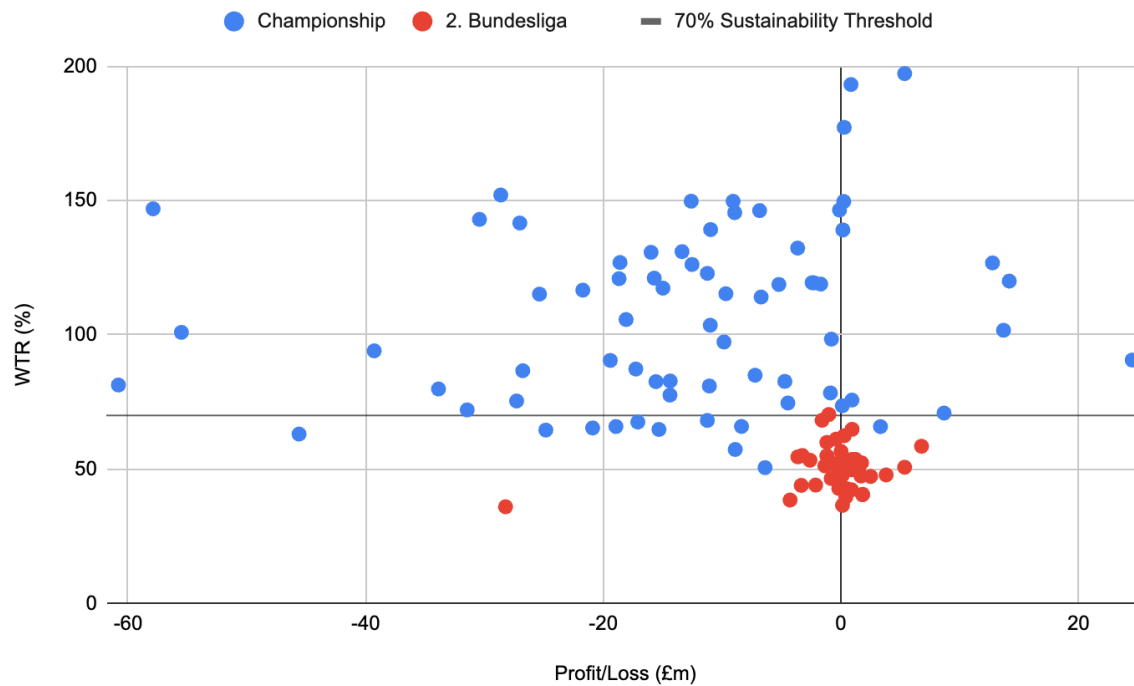
Figure 2: WTR vs Turnover for Championship and 2. Bundesliga Clubs



Source: Author's calculations using club financial statements.

Figure 3 extends this analysis by showing the relationship between WTR and net profit/loss. Championship clubs scatter across the top left corner, indicating that high wage expenditure is directly associated with financial losses. On the other hand, the 2. Bundesliga clubs cluster tightly in the lower-right area around the break-even point, with moderate WTRs. This relationship suggests that for Championship clubs, wage overspending is not simply a result of scale differences but reflects a deliberate, loss-generating financial strategy that is largely absent among German clubs. This divergence in financial culture motivates the project.

Figure 3: WTR vs Net Profit/Loss for Championship and 2. Bundesliga Clubs



Source: Author's calculations using club financial statements.

Table 3 summarises the number of gamblers and non-gamblers by proposition for Championship clubs over the three seasons from 2021/22 to 2023/24. Across all 53 club-season observations in the sample, 16 club-seasons are classified as gamblers, representing 30% of all observations. Of these 16, three are clubs in receipt of year-two parachute payments: Bournemouth (2021/22), Sheffield United (2022/23), and West Bromwich Albion (2022/23). Their inflated revenue bases make their classification as gamblers particularly noteworthy, as their wage growth still outpaces even their boosted revenue. This 30% gambling rate confirms that financially risky behaviour is a structural feature of the Championship, not an isolated occurrence.

Table 3: Financial Gambling by Proposition in the EFL Championship

Proposition	Championship position previous season	Seasons in study	Non-gamblers	Gamblers	Gamblers % of seasons in study
I	3rd to 6th*	9	3	6	67
II	7th to 12th	18	14	4	22
III	13th to 18th	18	14	4	22
IV	19th to 21st	8	6	2	25
Total		53	37	16	30

Table 4: Gambling Outcomes in the EFL Championship

Proposition	Championship position previous season	Outcome in current season						
		1st or 2nd	Play-off winner	3rd to 6th*	7th to 12th	13th to 18th	19th to 21st	22nd to 24th
I	3rd to 6th*	2	1			3		
II	7th to 12th			1	1	2		
III	13th to 18th			1	1	1		1
IV	19th to 21st				1	1		

Table 4a: Non-Gambling Outcomes in the EFL Championship

Proposition	Championship position previous season	Outcome in current season						
		1st or 2nd	Play-off winner	3rd to 6th*	7th to 12th	13th to 18th	19th to 21st	22nd to 24th
I	3rd to 6th*				2			1
II	7th to 12th			3	6	2	3	
III	13th to 18th			1	5	4	2	2
IV	19th to 21st			1		3		2

Source: Adapted from Evans et al. (2022); Author's calculations

The distribution across propositions is notably uneven and provides important insight into the motivations behind gambling behaviour. Proposition I clubs had the highest frequency of gambling, as 67% of club-seasons qualified as gambling according to the criteria defined above. Proposition I clubs face the highest marginal return from wage investment: promotion to the Premier League transforms a club's revenue base entirely. A club finishing fourth and spending aggressively on wages is, in effect, making a calculated bet that the payoff from promotion will far outweigh any short-term financial deterioration.

The comparison with non-gambling Proposition I clubs strongly supports this logic. Of the six gambling Proposition I club-seasons, three achieved promotion, while three regressed from their previous position to mid-table. The three non-gambling Proposition I club-seasons, by contrast, saw two clubs regress to mid-table and one club relegated. Every non-gambling Proposition I club-season then achieved, at best, the worst outcome faced by a gambling equivalent, with the largest downside embodied in a club being relegated to the EFL League One. This is the strongest takeaway from the Championship sample: among near-promotion clubs, gambling appears to be a precondition for actually achieving promotion from their position, as those that did not gamble universally regressed, and clubs that fail to exploit their prior position can even suffer relegation.

For Proposition II, the gambling rate was 22%. The results are mixed, as the distributions across both gambling and non-gambling club-seasons are similar, with no promotions or relegations recorded in either group. In this case, it would be irrational to gamble as there is clearly no meaningful difference in outcomes between clubs that gamble and do not gamble in Proposition II. Proposition III also has a gambling rate of 22%, and here the picture is similarly nuanced. As there are no promotions on either side of the gambling/non-gambling spectrum for this proposition, attention turns to the other end of the

table. For the four gambling Proposition III club-seasons, only one can be considered a success, as one club avoided relegation as its main objective. Two more teams avoided relegation, but did not advance to the automatic promotion spots, so it can only be considered partially successful, and there is one clear loser, with one club being relegated despite its financial gamble. Among non-gamblers in Proposition III, outcomes were similarly mixed: no promotions, two relegations, and the remainder scattered across mid-table, leaving the net benefit of gambling in this proposition inconclusive.

For Proposition IV, the gambling rate was 25%, and here, gambling proves successful across the sample. Both clubs that gambled in Proposition IV achieved their objective of relegation survival, where any finish outside the bottom three constitutes success. For non-gamblers, two of six club-seasons ended in relegation, with the other four being successful due to avoiding relegation. The finding here would suggest gambling to avoid relegation is a rational tactic, but one should be wary due to the very few instances of this in the studied sample.

The Championship results confirm that gambling is prevalent in the division, but most prevalent among clubs pursuing automatic promotion. Moreover, the comparative analysis of non-gambling results reveals that gambling is associated with better performance in almost all propositions. Looking at all 37 non-gambling clubs, none achieved a promotion and five relegations were recorded across all propositions. Looking at the 16 gambling instances, three were rewarded with promotion, and only one club was relegated. The strategy clearly carries downside risk for Proposition III clubs, where the one instance of a gamble resulting in relegation occurred, but that risk is not meaningfully higher than the relegation rate among non-gamblers in the same proposition. These findings diverge from Evans et al. (2022), where Proposition I clubs have the lowest proportion of gambling of all propositions, with

rates broadly uniform in the 41% to 47% range. The concentration of gambling at Proposition I in the current sample, and the decline in gambling across the board, may reflect evolving awareness of financial regulation, disruption of competitive balance driven by parachute payments, or the scale of financial burdens to owners becoming too severe.

Table 5 presents the corresponding results for the 2. Bundesliga over the same period. Across 35 club-seasons, only six are classified as gamblers, representing just 17% of observations. This project attributes the lower rate to the DFL's stricter licensing requirements and the near-universal 50+1 ownership structure, which constrains reckless financial activity.

Table 5: Financial Gambling by Proposition in the 2. Bundesliga

Proposition	2. Bundesliga position previous season	Seasons in study	Non-gamblers	Gamblers	Gamblers % of seasons in study
I	3rd to 6th*	11	9	2	18
II	7th to 11th	15	15	0	0
III	12th to 16th*	9	5	4	44
Total		35	29	6	17

Table 6: Gambling Outcomes in the 2. Bundesliga

Proposition	2. Bundesliga position previous season	Outcome in current season						
		1st or 2nd	3rd	4th to 6th	7th to 11th	12th to 15th	16th	17th to 18th
I	3rd to 6th*	1			1			
II	7th to 11th							
III	12th to 16th**				2	2		

Table 6a: Non-Gambling Outcomes in the 2. Bundesliga

Proposition	2. Bundesliga position previous season	Outcome in current season						
		1st or 2nd	3rd	4th to 6th	7th to 11th	12th to 15th	16th	17th to 18th
I	3rd to 6th*	1	3	2	2	1		
II	7th to 11th	1		7	4	3		
III	12th to 16th**				1	2		2

Source: Adapted from Evans et al. (2022); Author's calculations.

This breakdown reveals an interesting reversal of the trend observed in the Championship. Proposition I clubs in the 2. Bundesliga gamble at a rate of 18%, a substantially lower figure than the 67% recorded in the English context. Of the two gambling observations, one achieves automatic promotion, and one fails to improve its position. The nine non-gambling Proposition I clubs produce one automatic promotion, three finishes in third place, and five finishes in lower positions without relegation. Both gamblers and non-gamblers, therefore, produce one Zone A outcome each, and neither group records a relegation. Unlike the Championship, where gambling at near-promotion produces a clearly superior outcome profile to restraint, the evidence suggests that near-promotion clubs in the 2. Bundesliga do not gain a meaningful return from gambling, and those who exercise restraint achieve broadly equivalent outcomes. This underlines the structural point that with only three promotion places available in the 2. Bundesliga compared to six in the Championship, the probability-adjusted return on gambling is inherently lower regardless of a club's financial behaviour.

In Proposition II, none of the 15 mid-table club-seasons is classified as a gambler, the only proposition with a zero gambling rate across the entire sample. The 15 non-gambling

club-seasons produce one automatic promotion and 14 finishes scattered across mid-table, with no relegations recorded. The zero gambling rate suggests clubs are operating within regulatory and structural constraints that prevent wage escalation, rather than exercising restraint voluntarily. The distribution of outcomes from mid-table is wide and not meaningfully shaped by financial caution alone; clubs may also find the probability of promotion too low to justify the risk given their prior finish.

The gambling rate for clubs in Proposition III, the clubs in genuine relegation risk, is 44%. This is the most crucial finding for the German context since financial gambling within the 2. Bundesliga appears predominantly among clubs trying to avoid relegation instead of clubs seeking promotion. Table 6 shows that of the four Proposition III gamblers, all achieve their primary objective of avoiding relegation, in both the automatic and play-off sense. In contrast, Table 6a reveals that for the non-gamblers in Proposition III, there are two instances of automatic relegation alongside three clubs that survive relegation. This reveals a strategically rational response to relegation risk in the 2. Bundesliga, as every club that gambles to avoid relegation achieves its objective, while a substantial proportion of those who did not gamble are relegated.

The juxtaposition between Proposition I and Proposition III within the 2. Bundesliga also represents a significant empirical discovery. Promotion candidates are reluctant to gamble and produce outcomes broadly equivalent to non-gamblers, whereas relegation-threatened clubs engage in gambling at the highest frequency of any propositions and achieve results that compare favourably to their counterparts who did not gamble. Such a trend represents the inverse of the Championship, where gambling is concentrated among promotion candidates and yields stronger returns compared to non-gamblers across the same proposition. In the German context, financial gambling appears to be a strategy exclusive to

those clubs in relegation danger; adopted only when the stakes of relegation to the 3. Liga become sufficiently high, and the risk-reward calculation outweighs the structural and cultural forces that otherwise suppress it.

Discussion

The findings confirm the central hypothesis that financial gambling is more prevalent in the Championship than in the 2. Bundesliga. In addition, this research provides a more theoretically insightful finding, which suggests that the difference between the two leagues lies not only in the prevalence of financial gambling behaviour but also in its underlying motivation.

Overall, when looking at the aggregate level, the 30% gambling rate in the Championship out of 53 club-season observations is in stark contrast to the gambling rate of 17% in the 2. Bundesliga out of 35 observations. This disparity is consistent with the broader descriptive evidence from Figures 2 and 3, which make clear that Championship clubs regularly exceed the 70% WTR sustainability threshold, with some lower-income clubs even exceeding 150%, while all 2. Bundesliga clubs but one have ratios in the range of 40% to 65%. These figures alone suggest different financial cultures between the two leagues, independent of the league's revenue structure, club size, or outliers. Furthermore, the findings from the Championship are consistent with the results in the paper by Evans et al. (2022), wherein gambling is prevalent during the 2004-2019 sample period. This shows that financial risk-taking is a persistent part of the Championship landscape rather than a rare phenomenon.

Nevertheless, the most theoretically significant contribution of this project can be found in the distributional pattern of gambling across propositions in each division. In the Championship, gambling is most concentrated within the Proposition I clubs, which comprise clubs finishing third to sixth place in the previous season; 67% of club-seasons within this

proposition are observed to be gamblers. Economically speaking, the Proposition I clubs face the highest marginal benefit of gambling since promotion to the Premier League transforms a club's income base. Quinn (2025) estimates that Championship clubs not receiving parachute payments receive around £8 million in solidarity payments, compared with over £100 million in central distributions for the 20th-placed (last) Premier League club. Overspending on wages in pursuit of this reward is therefore economically rational despite the associated risks. That only two of six Proposition I gamblers achieve automatic promotion speaks to the strategy's uncertainty, not its irrationality..

Critically, this distributional pattern described above stands in contrast with Evans et al.'s (2022) earlier findings, where gambling is relatively equally distributed among the propositions, with Proposition I actually having the lowest gambling incidence. The high concentration of gambling in Proposition I within the current sample may reflect three compounding forces.

First, Premier League commercial revenue has grown substantially since 2019. Since the sample studied in Evans et al. (2022), the financial reward for promotion has grown significantly, owing primarily to a major new international TV rights deal introduced in the season 2022/23, which has seen a roughly 20% growth in international revenue equal payment per club in the Premier League (Premier League, 2019; 2024). Due to this, the expected value of promotion has significantly grown, making the gambling decision easier and more rational for clubs close to the promotion threshold.

Second, parachute payments are a primary driver of Championship clubs' willingness to gamble. As parachute payments are tied to the equally shared broadcasting revenue of the Premier League, when these revenue streams grow, the clubs in receipt of parachute payments benefit, while everyone else in the Championship suffers, acting as a competitive

disadvantage. This reasoning would point to both Proposition I clubs being aggressive gamblers in an attempt to compete with these parachute payment revenue-inflated clubs, but also to the clubs in Proposition II and III, who have seen a decline in gambling rates since Evans et al.'s (2022) study. When multiple clubs receiving parachute payments compete for the same promotion slots, a mid-table (Proposition II and III) club may reassess the probability-adjusted return on gambling and conclude that the odds of promotion are too slim to justify the financial gamble, given their position and the competition.

Third, the fear of falling further in the table may trigger clubs to double down on their approach. This dynamic reflects the sunk-cost fallacy. A club that ranks between third and sixth has typically incurred significant wage costs to reach that position, so owners and boards therefore face pressure to double down on their approach rather than to retrench. Cutting wage costs risks dropping the club into mid-table, thereby reducing commercial appeal, attendance rates, and player recruitment appeal. The fear of this downward spiral may make wage inflation seem less risky to owners than the financial consequences of retrenching. Queens Park Rangers and Reading both illustrate this dynamic in recent history, having built wage structures around parachute payments that subsequently expired, leaving them with considerably inflated wage bills relative to their income base and resulting in EFL financial settlements.

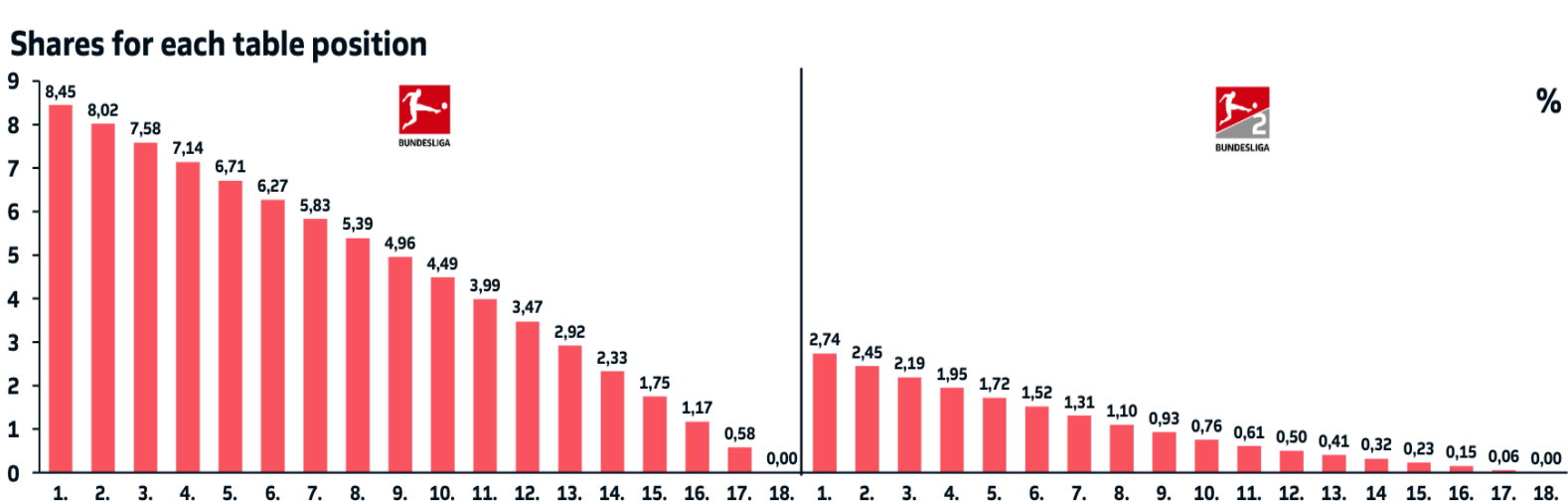
The 2. Bundesliga results provide an essentially inverse picture. Notably, none of the 15 Proposition II club-seasons is classified as a gambler. Mid-table 2. Bundesliga clubs appear universally unwilling to increase wage funding beyond natural revenue and profit growth to improve their performance in the hopes of achieving promotion. In the 2025/26 season, the reward of promotion is still substantial in the German context, with TV revenue distributions ranging from €30-80m per club annually in the top division compared to

€7-20m in the second division (Eckstein, 2025; Kryschak, 2025). Several structural and cultural forces explain this restraint, each reinforcing the others.

First, the DFL's licensing regime requires all clubs to participate in annual financial health checks and to demonstrate solvency before each season starts. Clubs cannot outspend their income base to the extent Championship clubs can.

Second, the revenue gap between the Bundesliga and the 2. Bundesliga is much narrower and smoother than in the English context. In terms of expected value of a promotion, there is far less incentive to risk lasting financial damage for the upside of promotion in Germany, as the revenue distributions are split over multiple seasons and assessed on different parameters, including national popularity, youth development, and a running sporting index that takes into account multiple season finishes. Figure 4 illustrates how this revenue sharing system affects each team across the top two divisions in Germany. As the DFL governs both divisions, it can smooth the financial transition between tiers. The Premier League, by contrast, has operated autonomously from the rest of English football since its formation in 1992. A Championship club can credibly project that a single promotion enduringly changes its financial horizon; the same cannot be said in Germany.

Figure 4: Shares of the national media revenue performance pillar by league position, showing the separate distributions for Bundesliga (left) and 2. Bundesliga (right)



Third, the league contains completely different incentive structures due to its promotion places. The Championship has two automatic promotion places at the top, and clubs finishing third to sixth have the opportunity to be promoted via the promotion play-offs. In the 2. Bundesliga, two teams at the top are automatically promoted, while only the third-placed team has a chance of promotion when they face the 16th-placed team from the Bundesliga. With fewer promotion places available, gambling is only rational if a club can credibly target a top-three finish.

Fourth, the near-universal 50+1 ownership model in the 2. Bundesliga prevents the type of owners and investors that can absorb financial losses that would otherwise typically be disciplined by market forces or governing structures. While member voting rights remove the moral hazard associated with affluent owners or investors, they introduce a new form of agency problem. Diffuse ownership structure and the emotional attachment of member-voters can still incentivise overspending, so the rule shifts the source of moral hazard rather than eradicating it. However, beyond the structural mechanics of the 50+1 structure, this ownership structure reflects a deeply embedded German footballing philosophy. The club is a community institution and source of pride rather than a vehicle for unconstrained sporting ambition. Attendance figures in the 2. Bundesliga are impressive, with average crowds exceeding 33,700 per match in 2024/25, comfortably exceeding the Championship and the top divisions of Italy and France, driving stable matchday revenue streams that reinforce each club's identity as a community institution (Bundesliga.com, 2025). This soft disciplinary mechanism from the 50+1 ownership model operates independently of formal DFL regulation and is arguably more culturally durable: it does not require enforcement because it is internalised by club leadership as a matter of institutional identity. In this sense, 50+1 does not merely remove the class of outside investors willing to fund wage speculation, but it

actively cultivates a boardroom culture in which such speculation would be considered a breach of the club's social contract with its members, regardless of whether the financial means existed.

Finally, the severity of relegation from the 2. Bundesliga to the 3. Liga fundamentally shapes the risk assessment for clubs. This existential threat simultaneously explains the suppressed level of promotion-focused gambles and the concentration of gambling among Proposition III clubs. Dropping into the 3. Liga is not just a minor setback, but can be a major structural shock for clubs. The 3. Liga operates entirely outside the DFL's central distribution framework, meaning relegation to that tier results in a dramatic collapse in central revenue, alongside the loss of the commercial and recruitment infrastructure that sustains second-division football. Recovery from this drop can take many seasons, and some clubs never return at all. Alemannia Aachen were relegated to the 3. Liga in 2012 and has spent over a decade attempting to return, while Dynamo Dresden's repeated yo-yo between the divisions illustrates the difficulty of establishing a sustainable foothold once financial stability is lost. This force acts as a gambling incentive and disincentive simultaneously. For clubs in genuine relegation danger, the existential severity of dropping to the next division justifies accepting short-term financial deterioration to fund squad improvement, which is reflected in the Proposition III gambling rate of 44%. However, for mid-table clubs, specifically Proposition II clubs, the same asymmetry acts as a brake on promotion-chasing risk: a miscalculated decision on wage expenditure that fails to deliver meaningful improvement in league position can, in the medium-term, erode financial resilience and hasten a descent toward the 3. Liga. Due to the condensed league format compared to the Championship, with only 18 clubs in the division, clubs play 34 fixtures compared to 46 in the Championship, leaving far less room to recover from a poor run of form. A sustained poor

run of form can prove decisive for a club in a season, with limited chances to recover league position given the shorter season.

Conclusion

This project evaluates whether Championship and 2. Bundesliga clubs engage in financial gambling, and whether the frequency and nature of that behaviour differs between the two divisions. The results support the central hypothesis. There is a greater prevalence of gambling in the Championship, and the distribution of clubs that gamble varies significantly between the two divisions.

The findings, with a 30% gambling rate in the Championship compared with only 17% in the 2. Bundesliga, confirm what is evident from the descriptive WTR analysis, which reveals that Championship clubs repeatedly exceed the 70% financial sustainability threshold, while effectively all 2. Bundesliga clubs operate well below it. The disparity in the financial culture of the two leagues is not coincidental. It reflects the considerably larger revenue gap between the Championship and the Premier League relative to its German equivalent, the distorting effect of parachute payments on competitive balance, the more varied promotion landscape of the Championship where six clubs have a realistic chance of promotion compared to three in Germany, and an ownership ecosystem in England that accommodates a broader class of investors willing to absorb financial losses in pursuit of sporting ambition.

This evidence has significant implications for the theory of regulatory legitimacy put forward by Evans et al. (2022). The Championship evidence demonstrates that the EFL Profitability and Sustainability rules still operate as a permission system rather than as a real constraint on finances, suggesting a structural issue rather than a cyclical one. The DFL's framework appears more effective, but the evidence cannot cleanly separate whether this is

attributable to the strictness of the licensing regime itself, the smaller financial incentive for promotion-chasing, or the cultural friction imposed by the 50+1 ownership model. These factors could be mutually reinforcing, providing an important message: financial prudence in the 2. Bundesliga may be the combined effect of multiple institutional factors rather than any single intervention.

Sustainability regulation implemented within a system where the expected value of promotion continues to grow will be absorbed by that system rather than disrupt it. Premier League broadcasting revenues and the compounding distortion of parachute payments ensure the underlying incentives remain overwhelming. Piecemeal rule-setting without addressing the underlying incentive architecture is likely to continue producing the pattern of regulatory legitimacy that this and prior research observes. The establishment of the Independent Football Regulator (IFR) in mid-2025 suggests financial distress has reached a critical threshold, and the British government has legislated an independent intervention to address this. The IFR will enforce compliance rules and lead a new licensing regime with the aim of safeguarding the footballing ecosystem against financial distress, poor governance, and threats to club heritage and fan engagement. It remains to be seen how effective this will be in practice, but it represents a step in the right direction; the question is whether the IFR will prove strict enough to deliver meaningful change.

For academic literature moving forward, this project opens several avenues. Extending the comparative framework to other European second divisions, such as France's Ligue 2, Spain's La Liga 2, or Italy's Serie B, would test whether the patterns observed here generalise beyond the English and German contexts, or are specific to their respective institutional environments. Incorporating transfer expenditure alongside wage data would enrich the gambling classifier, while extending the time horizon of the 2. Bundesliga sample

would allow a more robust assessment of whether the financial restraint observed here reflects a durable structural feature of German second-division football or a product of this specific period.

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